

Stakeholder Engagement Plan

Localized Agronomic Knowledge Retrieval in Nigeria

1. Stakeholder Groups

Farmers

Who they are: smallholder farmers across Nigeria's agro-ecological zones who submit natural language queries describing crop symptoms, pest damage, or soil conditions, ranging from routine agronomic questions to urgent issues such as disease outbreaks.

Why they matter to the project: they are the direct end users of the retrieval outcome. The entire purpose of the project, faster and more accurate access to agronomic knowledge, is measured by whether it actually reaches them with information they can act on. A system that scores well on benchmarks but surfaces irrelevant documents for an urgent disease query would fail the project's core goal even if the metric looks strong.

What matters most to them: getting relevant guidance quickly when a crop issue threatens their harvest or livelihood, having the system understand their own words and local terminology rather than only formal agronomic vocabulary, and trusting the information enough to act on it without needing to verify it elsewhere.

Agricultural Extension Officers

Who they are: the field staff who currently provide in-person agronomic advice, and who would use the retrieval system's output, or work alongside it, as a supplement to their own guidance once deployed.

Why they matter to the project: they are the operational users whose adoption determines whether the tool succeeds in practice. Their direct field exposure to farmer questions gives them the clearest view of where retrieval succeeds or fails, and their willingness to trust and recommend the tool directly affects whether it actually reduces the extension services bottleneck.

What matters most to them: a tool that extends their reach without replacing their judgment, especially in ambiguous or high-stakes cases such as a symptom that could indicate a fast-spreading disease. They want retrieved documents they can vouch for, an easy way to flag poor results, and confidence that the system supports rather than undermines their role as trusted local advisors.

2. Potential Benefits and Risks

Farmers

Benefits: faster access to relevant agronomic knowledge, less dependence on extension officer availability, and higher-stakes issues such as a spreading crop disease surfacing accurate documents sooner.

Harms and concerns: misretrieval of urgent queries, for example a disease symptom returning generic soil-health documents, causing real crop or income loss; unequal retrieval quality for farmers who describe symptoms in dialect or non-standard phrasing; and loss of trust if irrelevant or incorrect guidance goes unflagged.

Ethical, social, cultural, and economic considerations: fairness requires that retrieval quality not degrade based on regional dialect, literacy level, or how a query is phrased. Socially, there is a risk of eroding trust in digital agronomic tools if farmers feel their queries return unhelpful results. Culturally, the knowledge base may not fully represent regional farming practices or local names for crops, pests, and diseases across Nigeria's diverse agro-ecological zones. Economically, delayed or incorrect guidance on disease or pest issues carries direct financial risk through crop loss.

Agricultural Extension Officers

Benefits: less time spent manually searching reference material, more time for hands-on farmer support, and a tool that surfaces ambiguous or low-confidence queries for their review rather than silently guessing.

Harms and concerns: over-reliance on retrieved results leading to reduced scrutiny of the system's accuracy, erosion of independent diagnostic expertise over time, and concern that the tool signals a reduced need for their role rather than support for it.

Ethical, social, cultural, and economic considerations: responsibility for acting on retrieved guidance in high-stakes cases should remain shared between the system and the officer, not silently deferred to the tool. Officer buy-in and professional standing could be affected if the tool is rolled out without their input on what good retrieval looks like in the field. Reduced search time could be framed as freeing capacity for more farmer visits, or negatively as justification for reduced staffing, depending on how deployment is positioned.

3. Engagement Method

Farmers: Distributed Dialogue

Farmers are a large, dispersed population with brief, one-off contact with the system at the moment they submit a query, so a lightweight and asynchronous method fits this reality far better than a workshop or standing committee. It respects that most farmers want an answer rather than a seat on a panel, while still gathering signal on whether farmers who use non-standard or dialect phrasing are being underserved. In practice, a short and optional feedback prompt would appear after a query returns results, asking whether the farmer found the documents relevant and useful, alongside a low-friction way to flag results that missed the mark. Responses would be aggregated periodically, for example monthly, to identify patterns across many farmers rather than reviewing individual queries one by one.

Extension Officers: User Committee

Extension officers are a bounded, identifiable group who can be convened repeatedly over the life of the project, unlike farmers who interact with the system only briefly. Their sustained, hands-on exposure to farmer questions and local conditions makes them uniquely positioned to spot recurring retrieval failures, and a committee gives them real, ongoing influence over how the system evolves rather than having changes imposed on them. In practice, a small standing committee of extension officers, with rotating membership across regions to capture different agro-ecological zones, would meet regularly, for example biweekly or monthly, to review a sample of recent queries and retrieved documents, flag systematic gaps in the knowledge base, and discuss proposed changes to ranking or coverage before deployment. This formalizes expert review into a structured and accountable process with real decision input.

4. Influence on Project Design

Dataset and knowledge base: farmer distributed dialogue would surface query phrasing styles or regional terms that are underrepresented in the current knowledge base, informing targeted document collection. Extension officer feedback would identify systematically poor-performing query types, such as certain pest categories, guiding where additional curated documents are most needed.

Model and retrieval design: recurring committee review could inform decisions such as adjusting ranking thresholds for high-stakes query types, for example requiring higher relevance confidence before surfacing disease-related documents, or flagging low-confidence retrievals for mandatory officer review rather than direct farmer display.

Outputs: feedback could shape how results are presented, for example showing a confidence indicator or a short set of ranked options rather than a single forced top result, so officer or farmer judgment remains meaningfully part of the decision.

Evaluation: feedback from both groups would push evaluation beyond a single aggregate score toward per-region and per-query-type performance, ensuring the system is not judged successful while systematically underperforming for certain dialects, crops, or urgent categories.

Deployment decisions: concerns from both groups support a staged rollout, initially running the system in an assistive capacity that surfaces documents for extension officers to vet, rather than direct-to-farmer deployment for high-stakes categories, with broader farmer-facing access expanded only as trust and evidence of reliability accumulate.

Source note: the stakeholder analysis above draws on the project's stakeholder mapping notes and is grounded in the Agricultural Extension RAG Retrieval Benchmark used as this project's reference dataset.